Honolulu stress-test annex

Data as of 2026-10-28

Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list. Per-area counts from 1 to 9 are not shown. All figures are public Census Bureau, CDC and HHS estimates, not personal records.

Stress test, not a forecast. Each test changes one thing in the plan (a site closes, two hazards strike together, a road is cut, more people come) and counts who is affected, from the plan's own data. Deterministic code computes every number from the plan's data; people review the results. Results depend on the 30 sites opened in 2026; county-designated lists would change them.

People counted: residents of Census areas whose drive from the area's centre point to the nearest usable 2026 site was 15 minutes or less before the loss and is more than 15 minutes (or no route) after it; areas already beyond 15 minutes are not counted.

216 tests ran for Honolulu; 149 affect someone or need county data and are listed below. Two-site losses are listed only when they add people beyond either site alone. This is not a statement that the area has no shelter: counties designate and open shelters for each event, and many public buildings are not on the 2026 list.

Hurricane: top 10

1. **Lose George Fred Wright Wahiawā District Park and Mānoa Valley District Park (hurricane)** (test 2: Two sites lost)
   306,869 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 115; people more than losing either site alone: 91,451).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 115 · Drive limit (minutes): 15 · Island: Oʻahu · People more than losing either site alone: 91,451 · People left with no usable 2026 site on the island: 0 · Sites: George Fred Wright Wahiawā District Park, Mānoa Valley District Park
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
2. **Lose Kāneʻohe District Park and Mānoa Valley District Park (hurricane)** (test 2: Two sites lost)
   306,865 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 109; people more than losing either site alone: 91,447).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 109 · Drive limit (minutes): 15 · Island: Oʻahu · People more than losing either site alone: 91,447 · People left with no usable 2026 site on the island: 0 · Sites: Kāneʻohe District Park, Mānoa Valley District Park
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
3. **Lose Mānoa Valley District Park (hurricane)** (test 1: One site lost)
   215,418 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 85).
   Census areas pushed outside a 15-minute drive: 85 · Drive limit (minutes): 15 · Island: Oʻahu · May need a public shelter (likely): 6,462 · Site: Mānoa Valley District Park
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
4. **Stranded visitors for 3 or 7 days – Honolulu** (test 21: Stranded visitors)
   107,479 visitors on an average day (island level).
   By length of stay: 3 days: 644,874 meals, 322,437 gallons of water; 7 days: 1,504,706 meals, 752,353 gallons of water · Visitors on an average day (2025): 107,479
   Sources: DBEDT State of Hawaiʻi Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.

5. **Lose George Fred Wright Wahiawā District Park (hurricane)** (test 1: One site lost)
   91,451 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 30).

   Census areas pushed outside a 15-minute drive: 30 · Drive limit (minutes): 15 · Island: Oʻahu · May need a public shelter (likely): 2,744 · Site: George Fred Wright Wahiawā District Park

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

6. **All four counties at once (hurricane)** (test 9: Several counties at once)
   43,518 people who may need a public shelter (likely), counties together (cots: 43,518; shelter staff per shift: 2,921).

   Cots: 43,518 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 247; Chuukese: 58; Hawaiian: 28; Ilocano: 369; Japanese: 201; Korean: 99; Marshallese: 24; Pidgin (Hawaiʻi Creole English): 25; Spanish: 29; Tagalog/Filipino: 303; Vietnamese: 66 · Meals per day: 87,036 · People in shelter (likely): 43,518 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 2,921 · Water (gallons per day): 43,518

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

7. **Hurricane and inland flooding together – Honolulu** (test 5: Hurricane and inland flooding together)
   37,692 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 7; site gap: 144).

   Census areas: 322 · People if each hazard is counted separately: 1,379,565 · People in either footprint (counted once): 1,010,100 · May need a public shelter (likely): 37,692 · Site gap: 144 · Sites needed at 250 people per site (team assumption): 151 · 2026 sites usable for every hazard in the pair: 7

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250; A person in two footprints is counted once, at the higher rate: highest rate.

8. **Honolulu and Hawai‘i at once (hurricane)** (test 9: Several counties at once)
   36,368 people who may need a public shelter (likely), counties together (cots: 36,368; shelter staff per shift: 2,433).
   Cots: 36,368 · Counties: Honolulu, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 247; Chuukese: 58; Hawaiian: 23; Ilocano: 293; Japanese: 190; Korean: 99; Marshallese: 19; Pidgin (Hawai‘i Creole English): 25; Spanish: 15; Tagalog/Filipino: 261; Vietnamese: 66 · Meals per day: 72,736 · People in shelter (likely): 36,368 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 2,433 · Water (gallons per day): 36,368
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

9. **Stays of 3 to 30 days – Honolulu (hurricane)** (test 11: Longer stays)
   30,303 people who may need a public shelter (likely).
   By length of stay: 14 days: 1,212,120 sq ft of floor space, 848,484 meals, 1,516 toilets, 424,242 gallons of water; 3 days: 606,060 sq ft of floor space, 181,818 meals, 758 toilets, 90,909 gallons of water; 30 days: 1,212,120 sq ft of floor space, 1,818,180 meals, 1,516 toilets, 909,090 gallons of water; 7 days: 1,212,120 sq ft of floor space, 424,242 meals, 1,516 toilets, 212,121 gallons of water · People in shelter (likely): 30,303 · Per person per day: 2 meals; 1 gallons of water
   Sources: Florida Division of Emergency Management, Sheltering Guidance Aid (2010); CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Floor space and toilets use the 2-week factors for stays over 3 days, also past 14 days: 40 sq ft; 1 toilet per 20.

10. **Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites – O‘ahu: every potentially flooded highway segment closed** (test 24: Road closures (driving))
    23,903 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+98.8 min to any known site; +13.6 min to a 2026 site; cut off from every 2026 site: Hau‘ula · Census area 102.04, Nānākuli · Census area 9400.07).
    Census areas cut off: Hau‘ula · Census area 102.04, Nānākuli · Census area 9400.07 · Census areas cut off (over the limit or no route): 1 · Census areas cut off from every 2026 site: 2 · Census areas losing known sites within reach: 8 · Census areas with no route at all: 1 · Census areas routed: 170 · Census areas with a longer drive to a 2026 site: 3 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 98.8 · Most minutes added to the nearest 2026 site: 13.6 · Typical minutes added (median, where longer): 6.5 · People in areas cut off: 4,772 · People with a longer trip: 2,893 · People losing known sites within reach: 23,421 · Within reach means: 45-minute drive · Why this road:
    Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS); Hawai‘i Sea Level Rise Viewer, potential flooded highways at 3.2 ft (PacIOOS, via Hawai‘i Statewide GIS). Team assumptions: Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites: every potentially flooded highway segment on the island closed at once: all segments.

Wildfire: top 10

1. **Areas high in the State Fire Risk layer but not in the plan's wildfire areas, and the reverse – Honolulu** (test 23b: State Fire Risk Areas vs the plan's wildfire areas)

   236,444 people in areas rated high only by the State Fire Risk layer (Census areas high in the State layer only: 70; Census areas in the plan's wildfire areas only: 39).

   Areas high in both, of all areas high in either: 29% · Census areas high in both: 44 · Census areas in the plan's wildfire areas only: 39 · Census areas high in the State layer only: 70 · Largest plan-only areas: Urban Honolulu · Census area 67.04, Lāʻie · Census area 102.02, Kailua · Census area 111.06, ʻĀhuimanu / Kahaluʻu · Census area 103.05, Waimānalo · Census area 113 · People in plan-only areas: 141,501 · People in State-only areas: 236,444 · People in the plan's wildfire areas (FEMA National Risk Index): 314,263 · People in State High fire risk areas: 396,661 · Largest State-only areas: Māʻili · Census area 96.10, ʻEwa Beach · Census area 83.02, Waipahu · Census area 87.07, Mililani Mauka · Census area 89.29, ʻEwa Beach · Census area 84.18

   Sources: State Fire Risk Areas (DLNR Division of Forestry and Wildlife, via Hawaiʻi Statewide GIS); FEMA National Risk Index wildfire scores; CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks. Team assumptions: An area is 'high' in the State layer when at least half its people live in a High fire risk area: 50%.

2. **Stranded visitors for 3 or 7 days – Honolulu** (test 21: Stranded visitors)

   107,479 visitors on an average day (island level).

   By length of stay: 3 days: 644,874 meals, 322,437 gallons of water; 7 days: 1,504,706 meals, 752,353 gallons of water · Visitors on an average day (2025): 107,479

   Sources: DBEDT State of Hawaiʻi Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.

3. **Lose George Fred Wright Wahiawā District Park and Kāneʻohe District Park (wildfire)** (test 2: Two sites lost)

   66,396 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 17; people more than losing either site alone: 26,616).

   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 17 · Drive limit (minutes): 15 · Island: Oʻahu · People more than losing either site alone: 26,616 · People left with no usable 2026 site on the island: 0 · Sites: George Fred Wright Wahiawā District Park, Kāneʻohe District Park

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

4. **Lose Kāneʻohe District Park and Waiʻanae District Park (wildfire)** (test 2: Two sites lost)
   59,964 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 13; people more than losing either site alone: 20,184).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 13 · Drive limit (minutes): 15 · Island: Oʻahu · People more than losing either site alone: 20,184 · People left with no usable 2026 site on the island: 0 · Sites: Kāneʻohe District Park, Waiʻanae District Park
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
5. **Lose Kāneʻohe District Park (wildfire)** (test 1: One site lost)
   39,780 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 8).
   Census areas pushed outside a 15-minute drive: 8 · Drive limit (minutes): 15 · Island: Oʻahu · May need a public shelter (likely): 1,989 · Site: Kāneʻohe District Park
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
6. **Lose George Fred Wright Wahiawā District Park (wildfire)** (test 1: One site lost)
   26,616 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 9).
   Census areas pushed outside a 15-minute drive: 9 · Drive limit (minutes): 15 · Island: Oʻahu · May need a public shelter (likely): 1,331 · Site: George Fred Wright Wahiawā District Park
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

7. **Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites – O‘ahu: every potentially flooded highway segment closed** (test 24: Road closures (driving))
   23,903 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+98.8 min to any known site; +13.6 min to a 2026 site; cut off from every 2026 site: Hau‘ula · Census area 102.04, Nānākuli · Census area 9400.07).

   Census areas cut off: Hau‘ula · Census area 102.04, Nānākuli · Census area 9400.07 · Census areas cut off (over the limit or no route): 1 · Census areas cut off from every 2026 site: 2 · Census areas losing known sites within reach: 8 · Census areas with no route at all: 1 · Census areas routed: 170 · Census areas with a longer drive to a 2026 site: 3 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 98.8 · Most minutes added to the nearest 2026 site: 13.6 · Typical minutes added (median, where longer): 6.5 · People in areas cut off: 4,772 · People with a longer trip: 2,893 · People losing known sites within reach: 23,421 · Within reach means: 45-minute drive · Why this road:

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS); Hawai‘i Sea Level Rise Viewer, potential flooded highways at 3.2 ft (PacIOOS, via Hawai‘i Statewide GIS). Team assumptions: Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites: every potentially flooded highway segment on the island closed at once: all segments.

8. **All four counties at once (wildfire)** (test 9: Several counties at once)
   22,115 people who may need a public shelter (likely), counties together (cots: 22,115; shelter staff per shift: 1,534).

   Cots: 22,115 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 61; Chuukese: 14; Hawaiian: 9; Ilocano: 110; Japanese: 54; Korean: 24; Marshallese: 8; Pidgin (Hawai‘i Creole English): 10; Spanish: 12; Tagalog/Filipino: 84; Vietnamese: 16 · Meals per day: 44,229 · People in shelter (likely): 22,115 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 1,534 · Water (gallons per day): 22,115

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

9. **Honolulu and Hawai‘i at once (wildfire)** (test 9: Several counties at once)
   18,591 people who may need a public shelter (likely), counties together (cots: 18,591; shelter staff per shift: 1,285).
   Cots: 18,591 · Counties: Honolulu, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 61; Chuukese: 14; Hawaiian: 7; Ilocano: 77; Japanese: 49; Korean: 24; Marshallese: 6; Pidgin (Hawai‘i Creole English): 10; Spanish: 6; Tagalog/Filipino: 65; Vietnamese: 16 · Meals per day: 37,182 · People in shelter (likely): 18,591 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 1,285 · Water (gallons per day): 18,591
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.
10. **Kamehameha Highway (Route 83) at Waimea Bay, North Shore** (test 24: Road closures (driving))
    16,192 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+34.9 min to any known site; +0 min to a 2026 site).
    Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 6 · Census areas with no route at all: 0 · Census areas routed: 8 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 34.9 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 16,192 · Within reach means: 45-minute drive · Why this road: Single coastal road along the North Shore, exposed to high surf and flooding at Waimea (team assumption).
    Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

Tsunami: top 10

1. **Extreme tsunami zone (worst case) – Honolulu** (test 8: Extreme tsunami zone (worst case))
   208,201 people in the extreme tsunami evacuation zone (may need a public shelter (likely): 10,410; 2026 sites inside the extreme zone: 1).
   People in the extreme tsunami zone: 208,201 · People in the standard tsunami zone: 80,344 · May need a shelter (high): 22,902 · May need a public shelter (likely): 10,410 · May need a shelter (low): 4,164 · 2026 sites inside the extreme zone: 1 · Sites needed at 250 people per site (team assumption): 42 · 2026 sites usable outside both zones: 5
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; State tsunami evacuation zones and extreme tsunami evacuation zones (Hawai‘i Statewide GIS); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250.
2. **Visitors on the island(s) – Honolulu (tsunami)** (test 20: Visitors in tsunami zones)
   107,479 visitors on an average day (island level) (visitors inside the tsunami zone: needs county data (hotel and rental locations)).
   Detail: O‘ahu · Peak day: needs DBEDT monthly data (the Data Book gives the annual average) · Visitors on an average day (2025): 107,479 · Visitors inside the tsunami zone: needs county data (hotel and rental locations)
   Sources: DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census, 2025).
3. **Stranded visitors for 3 or 7 days – Honolulu** (test 21: Stranded visitors)
   107,479 visitors on an average day (island level).
   By length of stay: 3 days: 644,874 meals, 322,437 gallons of water; 7 days: 1,504,706 meals, 752,353 gallons of water · Visitors on an average day (2025): 107,479
   Sources: DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.
4. **Lose Kahuku Elementary School and Mānoa Valley District Park (tsunami)** (test 2: Two sites lost)
   25,180 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 20; people more than losing either site alone: 5,026).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 20 · Drive limit (minutes): 15 · Island: O‘ahu · People more than losing either site alone: 5,026 · People left with no usable 2026 site on the island: 0 · Sites: Kahuku Elementary School, Mānoa Valley District Park
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

5. **Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites – O‘ahu: every potentially flooded highway segment closed** (test 24: Road closures (driving))
   23,903 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+98.8 min to any known site; +13.6 min to a 2026 site; cut off from every 2026 site: Hau‘ula · Census area 102.04, Nānākuli · Census area 9400.07).
   Census areas cut off: Hau‘ula · Census area 102.04, Nānākuli · Census area 9400.07 · Census areas cut off (over the limit or no route): 1 · Census areas cut off from every 2026 site: 2 · Census areas losing known sites within reach: 8 · Census areas with no route at all: 1 · Census areas routed: 170 · Census areas with a longer drive to a 2026 site: 3 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 98.8 · Most minutes added to the nearest 2026 site: 13.6 · Typical minutes added (median, where longer): 6.5 · People in areas cut off: 4,772 · People with a longer trip: 2,893 · People losing known sites within reach: 23,421 · Within reach means: 45-minute drive · Why this road:
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS); Hawai‘i Sea Level Rise Viewer, potential flooded highways at 3.2 ft (PacIOOS, via Hawai‘i Statewide GIS). Team assumptions: Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites: every potentially flooded highway segment on the island closed at once: all segments.
6. **Lose Kāne‘ohe District Park and Mānoa Valley District Park (tsunami)** (test 2: Two sites lost)
   21,907 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 23; people more than losing either site alone: 1,753).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 23 · Drive limit (minutes): 15 · Island: O‘ahu · People more than losing either site alone: 1,753 · People left with no usable 2026 site on the island: 0 · Sites: Kāne‘ohe District Park, Mānoa Valley District Park
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
7. **Lose Mānoa Valley District Park (tsunami)** (test 1: One site lost)
   20,154 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 19).
   Census areas pushed outside a 15-minute drive: 19 · Drive limit (minutes): 15 · Island: O‘ahu · May need a public shelter (likely): 1,008 · Site: Mānoa Valley District Park
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

8. **Kamehameha Highway (Route 83) at Waimea Bay, North Shore** (test 24: Road closures (driving))
   16,192 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+34.9 min to any known site; +0 min to a 2026 site).

   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 6 · Census areas with no route at all: 0 · Census areas routed: 8 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 34.9 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 16,192 · Within reach means: 45-minute drive · Why this road: Single coastal road along the North Shore, exposed to high surf and flooding at Waimea (team assumption).

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

9. **Tsunami and coastal flooding together – Honolulu** (test 4: Tsunami and coastal flooding together)
   12,754 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 6; site gap: 46).

   Census areas: 106 · People if each hazard is counted separately: 323,750 · People in either footprint (counted once): 255,074 · May need a public shelter (likely): 12,754 · Site gap: 46 · Sites needed at 250 people per site (team assumption): 52 · 2026 sites usable for every hazard in the pair: 6

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions); State tsunami evacuation zones and extreme tsunami evacuation zones (Hawai‘i Statewide GIS). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250; A person in two footprints is counted once, at the higher rate: highest rate.

10. **All four counties at once (tsunami)** (test 9: Several counties at once)
    6,382 people who may need a public shelter (likely), counties together (cots: 6,382; shelter staff per shift: 442).

    Cots: 6,382 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 31; Chuukese: 8; Hawaiian: 5; Ilocano: 60; Japanese: 28; Korean: 13; Marshallese: 4; Pidgin (Hawai‘i Creole English): 4; Spanish: 7; Tagalog/Filipino: 46; Vietnamese: 9 · Meals per day: 12,764 · People in shelter (likely): 6,382 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 442 · Water (gallons per day): 6,382

    Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

Inland flooding: top 10

1. **Stranded visitors for 3 or 7 days – Honolulu** (test 21: Stranded visitors)
   107,479 visitors on an average day (island level).
   By length of stay: 3 days: 644,874 meals, 322,437 gallons of water; 7 days: 1,504,706 meals, 752,353 gallons of water · Visitors on an average day (2025): 107,479
   Sources: DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.
2. **Lose George Fred Wright Wahiawā District Park and Mānoa Valley District Park (inland flooding)** (test 2: Two sites lost)
   87,370 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 24; people more than losing either site alone: 31,380).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 24 · Drive limit (minutes): 15 · Island: O‘ahu · People more than losing either site alone: 31,380 · People left with no usable 2026 site on the island: 0 · Sites: George Fred Wright Wahiawā District Park, Mānoa Valley District Park
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
3. **Lose Kāne‘ohe District Park and Mānoa Valley District Park (inland flooding)** (test 2: Two sites lost)
   87,260 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 22; people more than losing either site alone: 31,270).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 22 · Drive limit (minutes): 15 · Island: O‘ahu · People more than losing either site alone: 31,270 · People left with no usable 2026 site on the island: 0 · Sites: Kāne‘ohe District Park, Mānoa Valley District Park
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
4. **Lose Mānoa Valley District Park (inland flooding)** (test 1: One site lost)
   55,990 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 16).
   Census areas pushed outside a 15-minute drive: 16 · Drive limit (minutes): 15 · Island: O‘ahu · May need a public shelter (likely): 2,800 · Site: Mānoa Valley District Park
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

5. **Hurricane and inland flooding together – Honolulu** (test 5: Hurricane and inland flooding together)
   37,692 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 7; site gap: 144).

   Census areas: 322 · People if each hazard is counted separately: 1,379,565 · People in either footprint (counted once): 1,010,100 · May need a public shelter (likely): 37,692 · Site gap: 144 · Sites needed at 250 people per site (team assumption): 151 · 2026 sites usable for every hazard in the pair: 7

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250; A person in two footprints is counted once, at the higher rate: highest rate.

6. **Lose George Fred Wright Wahiawā District Park (inland flooding)** (test 1: One site lost)
   31,380 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 8).

   Census areas pushed outside a 15-minute drive: 8 · Drive limit (minutes): 15 · Island: O‘ahu · May need a public shelter (likely): 1,569 · Site: George Fred Wright Wahiawā District Park

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

7. **All four counties at once (inland flooding)** (test 9: Several counties at once)
   25,452 people who may need a public shelter (likely), counties together (cots: 25,452; shelter staff per shift: 1,723).

   Cots: 25,452 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 100; Chuukese: 24; Hawaiian: 12; Ilocano: 153; Japanese: 82; Korean: 40; Marshallese: 10; Pidgin (Hawai‘i Creole English): 12; Spanish: 13; Tagalog/Filipino: 124; Vietnamese: 27 · Meals per day: 50,904 · People in shelter (likely): 25,452 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 1,723 · Water (gallons per day): 25,452

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

8. **Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites – O‘ahu: every potentially flooded highway segment closed** (test 24: Road closures (driving))
   23,903 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+98.8 min to any known site; +13.6 min to a 2026 site; cut off from every 2026 site: Hau‘ula · Census area 102.04, Nānākuli · Census area 9400.07).
   Census areas cut off: Hau‘ula · Census area 102.04, Nānākuli · Census area 9400.07 · Census areas cut off (over the limit or no route): 1 · Census areas cut off from every 2026 site: 2 · Census areas losing known sites within reach: 8 · Census areas with no route at all: 1 · Census areas routed: 170 · Census areas with a longer drive to a 2026 site: 3 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 98.8 · Most minutes added to the nearest 2026 site: 13.6 · Typical minutes added (median, where longer): 6.5 · People in areas cut off: 4,772 · People with a longer trip: 2,893 · People losing known sites within reach: 23,421 · Within reach means: 45-minute drive · Why this road:
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS); Hawai‘i Sea Level Rise Viewer, potential flooded highways at 3.2 ft (PacIOOS, via Hawai‘i Statewide GIS). Team assumptions: Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites: every potentially flooded highway segment on the island closed at once: all segments.
9. **Honolulu and Hawai‘i at once (inland flooding)** (test 9: Several counties at once)
   21,750 people who may need a public shelter (likely), counties together (cots: 21,750; shelter staff per shift: 1,468).
   Cots: 21,750 · Counties: Honolulu, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 100; Chuukese: 24; Hawaiian: 10; Ilocano: 120; Japanese: 78; Korean: 40; Marshallese: 8; Pidgin (Hawai‘i Creole English): 12; Spanish: 7; Tagalog/Filipino: 106; Vietnamese: 27 · Meals per day: 43,499 · People in shelter (likely): 21,750 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 1,468 · Water (gallons per day): 21,750
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.
10. **Dam failure and inland flooding together – Honolulu** (test 7: Dam failure and inland flooding together)
    19,378 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 7; site gap: 71).
    Census areas: 99 · People if each hazard is counted separately: 406,715 · People in either footprint (counted once): 387,551 · May need a public shelter (likely): 19,378 · Site gap: 71 · Sites needed at 250 people per site (team assumption): 78 · 2026 sites usable for every hazard in the pair: 7
    Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions); State of Hawai‘i dam evacuation zones (Hawai‘i Statewide GIS). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250; A person in two footprints is counted once, at the higher rate: highest rate; Dam failure uses the inland flooding shelter-seeking rate: 5%.

Coastal flooding: top 10

1. **Stranded visitors for 3 or 7 days – Honolulu** (test 21: Stranded visitors)
   107,479 visitors on an average day (island level).
   By length of stay: 3 days: 644,874 meals, 322,437 gallons of water; 7 days: 1,504,706 meals, 752,353 gallons of water · Visitors on an average day (2025): 107,479
   Sources: DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.
2. **Lose Kāne‘ohe District Park and Mānoa Valley District Park (coastal flooding)** (test 2: Two sites lost)
   81,819 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 33; people more than losing either site alone: 27,714).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 33 · Drive limit (minutes): 15 · Island: O‘ahu · People more than losing either site alone: 27,714 · People left with no usable 2026 site on the island: 0 · Sites: Kāne‘ohe District Park, Mānoa Valley District Park
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
3. **Lose Mānoa Valley District Park and Wai‘anae District Park (coastal flooding)** (test 2: Two sites lost)
   67,375 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 29; people more than losing either site alone: 13,270).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 29 · Drive limit (minutes): 15 · Island: O‘ahu · People more than losing either site alone: 13,270 · People left with no usable 2026 site on the island: 0 · Sites: Mānoa Valley District Park, Wai‘anae District Park
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
4. **Lose Mānoa Valley District Park (coastal flooding)** (test 1: One site lost)
   54,105 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 26).
   Census areas pushed outside a 15-minute drive: 26 · Drive limit (minutes): 15 · Island: O‘ahu · May need a public shelter (likely): 2,705 · Site: Mānoa Valley District Park
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

5. **Lose Kāneʻohe District Park (coastal flooding)** (test 1: One site lost)
   27,714 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 7).

   Census areas pushed outside a 15-minute drive: 7 · Drive limit (minutes): 15 · Island: Oʻahu · May need a public shelter (likely): 1,386 · Site: Kāneʻohe District Park

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

6. **Future sea-level-rise scenario (3.2 ft; State of Hawaiʻi SLR-XA), tested against 2026 sites – Oʻahu: every potentially flooded highway segment closed** (test 24: Road closures (driving))
   23,903 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+98.8 min to any known site; +13.6 min to a 2026 site; cut off from every 2026 site: Hauʻula · Census area 102.04, Nānākuli · Census area 9400.07).

   Census areas cut off: Hauʻula · Census area 102.04, Nānākuli · Census area 9400.07 · Census areas cut off (over the limit or no route): 1 · Census areas cut off from every 2026 site: 2 · Census areas losing known sites within reach: 8 · Census areas with no route at all: 1 · Census areas routed: 170 · Census areas with a longer drive to a 2026 site: 3 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 98.8 · Most minutes added to the nearest 2026 site: 13.6 · Typical minutes added (median, where longer): 6.5 · People in areas cut off: 4,772 · People with a longer trip: 2,893 · People losing known sites within reach: 23,421 · Within reach means: 45-minute drive · Why this road:

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawaiʻi DOT HPMS and surveyed mile markers (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Hawaiʻi DOE public schools (Hawaiʻi Statewide GIS); Hawaiʻi Sea Level Rise Viewer, potential flooded highways at 3.2 ft (PacIOOS, via Hawaiʻi Statewide GIS). Team assumptions: Future sea-level-rise scenario (3.2 ft; State of Hawaiʻi SLR-XA), tested against 2026 sites: every potentially flooded highway segment on the island closed at once: all segments.

7. **All four counties at once (coastal flooding)** (test 9: Several counties at once)
   17,662 people who may need a public shelter (likely), counties together (cots: 17,662; shelter staff per shift: 1,171).

   Cots: 17,662 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 99; Chuukese: 23; Hawaiian: 13; Ilocano: 154; Japanese: 83; Korean: 39; Marshallese: 11; Pidgin (Hawaiʻi Creole English): 15; Spanish: 15; Tagalog/Filipino: 122; Vietnamese: 26 · Meals per day: 35,324 · People in shelter (likely): 17,662 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 1,171 · Water (gallons per day): 17,662

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

8. **Kamehameha Highway (Route 83) at Waimea Bay, North Shore** (test 24: Road closures (driving))
   16,192 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+34.9 min to any known site; +0 min to a 2026 site).

   Census areas cut off: none · Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 6 · Census areas with no route at all: 0 · Census areas routed: 8 · Census areas with a longer drive to a 2026 site: 0 · Cut off means: over 45 minutes or no route · Most minutes added to any known site: 34.9 · Most minutes added to the nearest 2026 site: 0 · Typical minutes added (median, where longer): 0 · People in areas cut off: 0 · People with a longer trip: 0 · People losing known sites within reach: 16,192 · Within reach means: 45-minute drive · Why this road: Single coastal road along the North Shore, exposed to high surf and flooding at Waimea (team assumption).

   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.

9. **Honolulu and Hawai‘i at once (coastal flooding)** (test 9: Several counties at once)
   14,869 people who may need a public shelter (likely), counties together (cots: 14,869; shelter staff per shift: 977).

   Cots: 14,869 · Counties: Honolulu, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 99; Chuukese: 23; Hawaiian: 11; Ilocano: 122; Japanese: 78; Korean: 39; Marshallese: 9; Pidgin (Hawai‘i Creole English): 15; Spanish: 9; Tagalog/Filipino: 105; Vietnamese: 26 · Meals per day: 29,738 · People in shelter (likely): 14,869 · Rounding: counties' unrounded numbers added, then rounded up once · Shelter staff per shift: 977 · Water (gallons per day): 14,869

   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: One rounding rule for counties together: add the unrounded county numbers, then round up once: round up.

10. **Tsunami and coastal flooding together – Honolulu** (test 4: Tsunami and coastal flooding together)
    12,754 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 6; site gap: 46).

    Census areas: 106 · People if each hazard is counted separately: 323,750 · People in either footprint (counted once): 255,074 · May need a public shelter (likely): 12,754 · Site gap: 46 · Sites needed at 250 people per site (team assumption): 52 · 2026 sites usable for every hazard in the pair: 6

    Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions); State tsunami evacuation zones and extreme tsunami evacuation zones (Hawai‘i Statewide GIS). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250; A person in two footprints is counted once, at the higher rate: highest rate.

Every result, ranked by people affected

Rank	Test	Scenario	Hazard	Result
1	2	Lose George Fred Wright Wahiawā District Park and Mānoa Valley District Park (hurricane)	Hurricane	306,869 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 115; people more than losing either site alone: 91,451)
1	9	All four counties at once (hurricane)	Hurricane	43,518 people who may need a public shelter (likely), counties together (cots: 43,518; shelter staff per shift: 2,921)
2	2	Lose Kāneʻohe District Park and Mānoa Valley District Park (hurricane)	Hurricane	306,865 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 109; people more than losing either site alone: 91,447)
2	9	Honolulu and Hawaiʻi at once (hurricane)	Hurricane	36,368 people who may need a public shelter (likely), counties together (cots: 36,368; shelter staff per shift: 2,433)
3	23b	Areas high in the State Fire Risk layer but not in the plan's wildfire areas, and the reverse – Honolulu	Wildfire	236,444 people in areas rated high only by the State Fire Risk layer (Census areas high in the State layer only: 70; Census areas in the plan's wildfire areas only: 39)
3	9	Honolulu and Maui at once (hurricane)	Hurricane	35,248 people who may need a public shelter (likely), counties together (cots: 35,248; shelter staff per shift: 2,357)
4	2	Lose Mānoa Valley District Park and Waiʻanae District Park (hurricane)	Hurricane	235,602 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 90; people more than losing either site alone: 20,184)
4	9	Kauaʻi and Honolulu at once (hurricane)	Hurricane	32,509 people who may need a public shelter (likely), counties together (cots: 32,509; shelter staff per shift: 2,177)

Rank	Test	Scenario	Hazard	Result
5	2	Lose Kahuku Elementary School and Mānoa Valley District Park (hurricane)	Hurricane	222,398 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 86; people more than losing either site alone: 6,980)
5	9	All four counties at once (inland flooding)	Inland flooding	25,452 people who may need a public shelter (likely), counties together (cots: 25,452; shelter staff per shift: 1,723)
6	1	Lose Mānoa Valley District Park (hurricane)	Hurricane	215,418 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 85)
6	9	All four counties at once (wildfire)	Wildfire	22,115 people who may need a public shelter (likely), counties together (cots: 22,115; shelter staff per shift: 1,534)
7	9	Honolulu and Hawai‘i at once (inland flooding)	Inland flooding	21,750 people who may need a public shelter (likely), counties together (cots: 21,750; shelter staff per shift: 1,468)
8	9	Honolulu and Maui at once (inland flooding)	Inland flooding	21,106 people who may need a public shelter (likely), counties together (cots: 21,106; shelter staff per shift: 1,423)
9	8	Extreme tsunami zone (worst case) – Honolulu	Tsunami	208,201 people in the extreme tsunami evacuation zone (may need a public shelter (likely): 10,410; 2026 sites inside the extreme zone: 1)
9	9	Kaua‘i and Honolulu at once (inland flooding)	Inland flooding	19,544 people who may need a public shelter (likely), counties together (cots: 19,544; shelter staff per shift: 1,322)
10	2	Lose George Fred Wright Wahiawā District Park and Kāne‘ohe District Park (hurricane)	Hurricane	182,898 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 54; people more than losing either site alone: 91,447)

Rank	Test	Scenario	Hazard	Result
10	9	Honolulu and Hawai‘i at once (wildfire)	Wildfire	18,591 people who may need a public shelter (likely), counties together (cots: 18,591; shelter staff per shift: 1,285)
11	2	Lose George Fred Wright Wahiawā District Park and Wai‘anae District Park (hurricane)	Hurricane	111,635 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 35; people more than losing either site alone: 20,184)
11	9	Honolulu and Maui at once (wildfire)	Wildfire	18,155 people who may need a public shelter (likely), counties together (cots: 18,155; shelter staff per shift: 1,258)
12	2	Lose Kāne‘ohe District Park and Wai‘anae District Park (hurricane)	Hurricane	111,631 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 29; people more than losing either site alone: 20,184)
12	9	All four counties at once (coastal flooding)	Coastal flooding	17,662 people who may need a public shelter (likely), counties together (cots: 17,662; shelter staff per shift: 1,171)
13	20	Visitors on the island(s) – Honolulu (tsunami)	Tsunami	107,479 visitors on an average day (island level) (visitors inside the tsunami zone: needs county data (hotel and rental locations))
13	9	Kaua‘i and Honolulu at once (wildfire)	Wildfire	16,795 people who may need a public shelter (likely), counties together (cots: 16,795; shelter staff per shift: 1,163)
14	21	Stranded visitors for 3 or 7 days – Honolulu	All hazards	107,479 visitors on an average day (island level)
14	9	Honolulu and Hawai‘i at once (coastal flooding)	Coastal flooding	14,869 people who may need a public shelter (likely), counties together (cots: 14,869; shelter staff per shift: 977)

Rank	Test	Scenario	Hazard	Result
15	2	Lose George Fred Wright Wahiawā District Park and Kahuku Elementary School (hurricane)	Hurricane	98,431 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 31; people more than losing either site alone: 6,980)
15	9	Honolulu and Maui at once (coastal flooding)	Coastal flooding	14,005 people who may need a public shelter (likely), counties together (cots: 14,005; shelter staff per shift: 926)
16	2	Lose Kahuku Elementary School and Kāneʻohe District Park (hurricane)	Hurricane	98,427 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 25; people more than losing either site alone: 6,980)
16	9	Kauaʻi and Honolulu at once (coastal flooding)	Coastal flooding	13,130 people who may need a public shelter (likely), counties together (cots: 13,130; shelter staff per shift: 868)
17	1	Lose George Fred Wright Wahiawā District Park (hurricane)	Hurricane	91,451 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 30)
20	1	Lose Kāneʻohe District Park (hurricane)	Hurricane	91,447 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 24)
20	9	All four counties at once (tsunami)	Tsunami	6,382 people who may need a public shelter (likely), counties together (cots: 6,382; shelter staff per shift: 442)
22	9	Honolulu and Maui at once (tsunami)	Tsunami	5,326 people who may need a public shelter (likely), counties together (cots: 5,326; shelter staff per shift: 362)
23	2	Lose George Fred Wright Wahiawā District Park and Mānoa Valley District Park (inland flooding)	Inland flooding	87,370 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 24; people more than losing either site alone: 31,380)

Rank	Test	Scenario	Hazard	Result
24	2	Lose Kāneʻohe District Park and Mānoa Valley District Park (inland flooding)	Inland flooding	87,260 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 22; people more than losing either site alone: 31,270)
24	9	Honolulu and Hawaiʻi at once (tsunami)	Tsunami	4,623 people who may need a public shelter (likely), counties together (cots: 4,623; shelter staff per shift: 316)
25	2	Lose Kāneʻohe District Park and Mānoa Valley District Park (coastal flooding)	Coastal flooding	81,819 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 33; people more than losing either site alone: 27,714)
26	2	Lose Mānoa Valley District Park and Waiʻanae District Park (coastal flooding)	Coastal flooding	67,375 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 29; people more than losing either site alone: 13,270)
26	9	Kauaʻi and Honolulu at once (tsunami)	Tsunami	4,469 people who may need a public shelter (likely), counties together (cots: 4,469; shelter staff per shift: 306)
27	2	Lose George Fred Wright Wahiawā District Park and Kāneʻohe District Park (wildfire)	Wildfire	66,396 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 17; people more than losing either site alone: 26,616)
28	2	Lose Mānoa Valley District Park and Waiʻanae District Park (inland flooding)	Inland flooding	65,936 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 18; people more than losing either site alone: 9,946)

Rank	Test	Scenario	Hazard	Result
29	2	Lose Kahuku Elementary School and Mānoa Valley District Park (inland flooding)	Inland flooding	62,970 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 17; people more than losing either site alone: 6,980)
30	2	Lose George Fred Wright Wahiawā District Park and Kāneʻohe District Park (inland flooding)	Inland flooding	62,650 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 14; people more than losing either site alone: 31,270)
31	2	Lose Kahuku Elementary School and Mānoa Valley District Park (coastal flooding)	Coastal flooding	61,085 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 27; people more than losing either site alone: 6,980)
32	2	Lose Kāneʻohe District Park and Waiʻanae District Park (wildfire)	Wildfire	59,964 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 13; people more than losing either site alone: 20,184)
33	2	Lose Kāneʻohe District Park and Mānoa Valley District Park (wildfire)	Wildfire	57,164 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 15; people more than losing either site alone: 17,384)
34	1	Lose Mānoa Valley District Park (inland flooding)	Inland flooding	55,990 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 16)
37	1	Lose Mānoa Valley District Park (coastal flooding)	Coastal flooding	54,105 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 26)

Rank	Test	Scenario	Hazard	Result
41	2	Lose George Fred Wright Wahiawā District Park and Waiʻanae District Park (wildfire)	Wildfire	46,800 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 14; people more than losing either site alone: 20,184)
42	2	Lose Kahuku Elementary School and Kāneʻohe District Park (wildfire)	Wildfire	46,760 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 9; people more than losing either site alone: 6,980)
43	2	Lose George Fred Wright Wahiawā District Park and Mānoa Valley District Park (wildfire)	Wildfire	44,000 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 16; people more than losing either site alone: 17,384)
44	2	Lose George Fred Wright Wahiawā District Park and Waiʻanae District Park (inland flooding)	Inland flooding	41,326 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 10; people more than losing either site alone: 9,946)
45	2	Lose Kāneʻohe District Park and Waiʻanae District Park (inland flooding)	Inland flooding	41,216 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 8; people more than losing either site alone: 9,946)
46	2	Lose Kāneʻohe District Park and Waiʻanae District Park (coastal flooding)	Coastal flooding	40,984 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 10; people more than losing either site alone: 13,270)
47	1	Lose Kāneʻohe District Park (wildfire)	Wildfire	39,780 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 8)

Rank	Test	Scenario	Hazard	Result
50	2	Lose George Fred Wright Wahiawā District Park and Kahuku Elementary School (inland flooding)	Inland flooding	38,360 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 9; people more than losing either site alone: 6,980)
51	2	Lose Kahuku Elementary School and Kāneʻohe District Park (inland flooding)	Inland flooding	38,250 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 7; people more than losing either site alone: 6,980)
52	5	Hurricane and inland flooding together – Honolulu	Hurricane + Inland flooding	37,692 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 7; site gap: 144)
53	2	Lose Mānoa Valley District Park and Waiʻanae District Park (wildfire)	Wildfire	37,568 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 12; people more than losing either site alone: 17,384)
54	2	Lose Kahuku Elementary School and Kāneʻohe District Park (coastal flooding)	Coastal flooding	34,694 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 8; people more than losing either site alone: 6,980)
55	2	Lose George Fred Wright Wahiawā District Park and Kahuku Elementary School (wildfire)	Wildfire	33,596 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 10; people more than losing either site alone: 6,980)
56	1	Lose George Fred Wright Wahiawā District Park (inland flooding)	Inland flooding	31,380 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 8)
59	1	Lose Kāneʻohe District Park (inland flooding)	Inland flooding	31,270 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 6)

Rank	Test	Scenario	Hazard	Result
62	11	Stays of 3 to 30 days – Honolulu (hurricane)	Hurricane	30,303 people who may need a public shelter (likely)
63	1	Lose Kāneʻohe District Park (coastal flooding)	Coastal flooding	27,714 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 7)
67	2	Lose Kahuku Elementary School and Waiʻanae District Park (hurricane)	Hurricane	27,164 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 6; people more than losing either site alone: 6,980)
68	2	Lose Kahuku Elementary School and Waiʻanae District Park (wildfire)	Wildfire	27,164 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 6; people more than losing either site alone: 6,980)
69	1	Lose George Fred Wright Wahiawā District Park (wildfire)	Wildfire	26,616 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 9)
72	2	Lose Nānākuli Elementary School and Waiʻanae District Park (hurricane)	Hurricane	26,080 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 6; people more than losing either site alone: 5,896)
73	2	Lose Nānākuli Elementary School and Waiʻanae District Park (wildfire)	Wildfire	26,080 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 6; people more than losing either site alone: 5,896)
74	2	Lose Kahuku Elementary School and Mānoa Valley District Park (tsunami)	Tsunami	25,180 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 20; people more than losing either site alone: 5,026)

Rank	Test	Scenario	Hazard	Result
75	2	Lose Kahuku Elementary School and Mānoa Valley District Park (wildfire)	Wildfire	24,364 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 8; people more than losing either site alone: 6,980)
76	24	Future sea-level-rise scenario (3.2 ft; State of Hawai‘i SLR-XA), tested against 2026 sites – O‘ahu: every potentially flooded highway segment closed	All hazards	23,903 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+98.8 min to any known site; +13.6 min to a 2026 site; cut off from every 2026 site: Hau‘ula · Census area 102.04, Nānākuli · Census area 9400.07)
77	2	Lose Kāne‘ohe District Park and Mānoa Valley District Park (tsunami)	Tsunami	21,907 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 23; people more than losing either site alone: 1,753)
78	2	Lose Kahuku Elementary School and Wai‘anae District Park (coastal flooding)	Coastal flooding	20,250 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4; people more than losing either site alone: 6,980)
79	10	Rate at which demand needs more sites than are usable – Honolulu (hurricane)	Hurricane	20,202 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 2%; usable 2026 sites: 7)
80	1	Lose Wai‘anae District Park (hurricane)	Hurricane	20,184 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 5)
81	1	Lose Wai‘anae District Park (wildfire)	Wildfire	20,184 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 5)
84	1	Lose Mānoa Valley District Park (tsunami)	Tsunami	20,154 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 19)

Rank	Test	Scenario	Hazard	Result
88	7	Dam failure and inland flooding together – Honolulu	Dam failure + Inland flooding	19,378 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 7; site gap: 71)
89	11	Stays of 3 to 30 days – Honolulu (inland flooding)	Inland flooding	18,473 people who may need a public shelter (likely)
90	1	Lose Mānoa Valley District Park (wildfire)	Wildfire	17,384 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 7)
93	2	Lose Kahuku Elementary School and Waiʻanae District Park (inland flooding)	Inland flooding	16,926 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 6,980)
94	24	Kamehameha Highway (Route 83) at Waimea Bay, North Shore	All hazards	16,192 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (+34.9 min to any known site; +0 min to a 2026 site)
95	2	Lose Nānākuli Elementary School and Waiʻanae District Park (inland flooding)	Inland flooding	15,842 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3; people more than losing either site alone: 5,896)
96	6	Wildfire and power loss together – Honolulu	Wildfire + Power loss	15,713 people who may need a public shelter (likely) (outlets per day if the power is out: 1,893)
97	11	Stays of 3 to 30 days – Honolulu (wildfire)	Wildfire	15,713 people who may need a public shelter (likely)
98	2	Lose Nānākuli Elementary School and Nānākuli High and Intermediate School (hurricane)	Hurricane	13,545 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 5; people more than losing either site alone: 13,545)

Rank	Test	Scenario	Hazard	Result
99	2	Lose Nānākuli Elementary School and Nānākuli High and Intermediate School (wildfire)	Wildfire	13,545 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 5; people more than losing either site alone: 13,545)
100	1	Lose Waiʻanae District Park (coastal flooding)	Coastal flooding	13,270 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 3)
104	4	Tsunami and coastal flooding together – Honolulu	Tsunami + Coastal flooding	12,754 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 6; site gap: 46)
105	13	Kūpuna living alone, no car, far from a site – Honolulu (hurricane)	Hurricane	12,357 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 176)
106	11	Stays of 3 to 30 days – Honolulu (coastal flooding)	Coastal flooding	12,170 people who may need a public shelter (likely)
107	2	Lose Nānākuli Elementary School and Nānākuli High and Intermediate School (tsunami)	Tsunami	10,894 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 5; people more than losing either site alone: 10,894)
108	1	Lose Waiʻanae District Park (inland flooding)	Inland flooding	9,946 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2)
110	10	Rate at which demand needs more sites than are usable – Honolulu (inland flooding)	Inland flooding	7,389 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 2%; usable 2026 sites: 7)
111	1	Lose Kahuku Elementary School (coastal flooding)	Coastal flooding	6,980 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
112	1	Lose Kahuku Elementary School (hurricane)	Hurricane	6,980 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)

Rank	Test	Scenario	Hazard	Result
113	1	Lose Kahuku Elementary School (inland flooding)	Inland flooding	6,980 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
114	1	Lose Kahuku Elementary School (wildfire)	Wildfire	6,980 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)
124	2	Lose Kahuku Elementary School and Kāneʻohe District Park (tsunami)	Tsunami	6,778 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 5; people more than losing either site alone: 1,753)
125	12	Power out 1 to 14 days – Honolulu (hurricane)	Hurricane	6,385 power-dependent residents in the risk area (outlets per day while the power is out: 6,385)
126	10	Rate at which demand needs more sites than are usable – Honolulu (wildfire)	Wildfire	6,285 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 2%; usable 2026 sites: 7)
127	3	Treat DOE schools in FEMA zone D as not usable (hurricane)	Hurricane	5,752 people in the risk area pushed outside a 15-minute drive to a usable site or candidate school (Census areas pushed outside a 15-minute drive: 3)
128	23a	No car, inside the tsunami zone, over 15 minutes' walk to a safe zone – Honolulu	Tsunami	5,298 people without a car more than 15 minutes' walk from a tsunami safe zone (Census areas over 15 minutes' walk: 23; longest walk (minutes): 31.1)
129	2	Lose Nānākuli Elementary School and Nānākuli High and Intermediate School (inland flooding)	Inland flooding	5,192 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1; people more than losing either site alone: 5,192)
130	1	Lose Kahuku Elementary School (tsunami)	Tsunami	5,026 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1)

Rank	Test	Scenario	Hazard	Result
134	16	Chuukese: reliable count, 0 of 15 messages translated – Honolulu	All hazards	4,948 people who speak English less than very well (messages translated: 0)
135	13	Kūpuna living alone, no car, far from a site – Honolulu (inland flooding)	Inland flooding	4,897 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 46)
136	10	Rate at which demand needs more sites than are usable – Honolulu (coastal flooding)	Coastal flooding	4,868 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 2%; usable 2026 sites: 7)
137	18	Add the 2026 Point-in-Time count – Honolulu (coastal flooding)	Coastal flooding	4,539 people added by the 2026 Point-in-Time count (increase over the likely demand: 37.3%)
138	18	Add the 2026 Point-in-Time count – Honolulu (hurricane)	Hurricane	4,539 people added by the 2026 Point-in-Time count (increase over the likely demand: 15%)
139	18	Add the 2026 Point-in-Time count – Honolulu (inland flooding)	Inland flooding	4,539 people added by the 2026 Point-in-Time count (increase over the likely demand: 24.6%)
140	18	Add the 2026 Point-in-Time count – Honolulu (tsunami)	Tsunami	4,539 people added by the 2026 Point-in-Time count (increase over the likely demand: 113%)
141	18	Add the 2026 Point-in-Time count – Honolulu (wildfire)	Wildfire	4,539 people added by the 2026 Point-in-Time count (increase over the likely demand: 28.9%)
142	13	Kūpuna living alone, no car, far from a site – Honolulu (coastal flooding)	Coastal flooding	4,238 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 45)
143	11	Stays of 3 to 30 days – Honolulu (tsunami)	Tsunami	4,017 people who may need a public shelter (likely)
144	13	Kūpuna living alone, no car, far from a site – Honolulu (wildfire)	Wildfire	3,815 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 46)
145	15	Power-dependent and far from a site – Honolulu (hurricane)	Hurricane	3,303 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 176)

Rank	Test	Scenario	Hazard	Result
146	3	Treat DOE schools in FEMA zone D as not usable (wildfire)	Wildfire	2,361 people in the risk area pushed outside a 15-minute drive to a usable site or candidate school (Census areas pushed outside a 15-minute drive: 1)
147	12	Power out 1 to 14 days – Honolulu (inland flooding)	Inland flooding	2,315 power-dependent residents in the risk area (outlets per day while the power is out: 2,315)
148	12	Power out 1 to 14 days – Honolulu (wildfire)	Wildfire	1,893 power-dependent residents in the risk area (outlets per day while the power is out: 1,893)
149	25	Kamehameha Highway (Route 83) at Waimea Bay, North Shore – walking	All hazards	1,787 people in areas with a longer walk to a 2026 site or cut off on foot (+100.1 min to any known site; +100.1 min to a 2026 site)
150	1	Lose Kāneʻohe District Park (tsunami)	Tsunami	1,753 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 4)
154	10	Rate at which demand needs more sites than are usable – Honolulu (tsunami)	Tsunami	1,607 people who may need a shelter at the rate where usable sites run out (rate when usable sites run out: 2%; usable 2026 sites: 6)
155	12	Power out 1 to 14 days – Honolulu (coastal flooding)	Coastal flooding	1,552 power-dependent residents in the risk area (outlets per day while the power is out: 1,552)
156	13	Kūpuna living alone, no car, far from a site – Honolulu (tsunami)	Tsunami	1,490 kūpuna living alone without a car, outside a 15-minute drive (upper bound) (Census areas outside a 15-minute drive: 52)
157	14	Hearing difficulty and outside siren range (tsunami, at night) – Honolulu	Tsunami	1,366 people with hearing difficulty outside rated siren range (upper bound)
158	15	Power-dependent and far from a site – Honolulu (inland flooding)	Inland flooding	1,255 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 46)
159	15	Power-dependent and far from a site – Honolulu (wildfire)	Wildfire	1,100 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 46)

Rank	Test	Scenario	Hazard	Result
160	16	Marshallese: reliable count, 0 of 15 messages translated – Honolulu	All hazards	1,021 people who speak English less than very well (messages translated: 0)
161	15	Power-dependent and far from a site – Honolulu (coastal flooding)	Coastal flooding	832 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 45)
162	3	Treat DOE schools in FEMA zone D as not usable (tsunami)	Tsunami	824 people in the risk area pushed outside a 15-minute drive to a usable site or candidate school (Census areas pushed outside a 15-minute drive: 3)
163	12	Power out 1 to 14 days – Honolulu (tsunami)	Tsunami	507 power-dependent residents in the risk area (outlets per day while the power is out: 507)
164	15	Power-dependent and far from a site – Honolulu (tsunami)	Tsunami	271 power-dependent residents outside a 15-minute drive (Census areas outside a 15-minute drive: 52)
184	17	Pet households vs pet-friendly sites – Honolulu (hurricane)	Hurricane	7,068 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 7)
185	17	Pet households vs pet-friendly sites – Honolulu (inland flooding)	Inland flooding	4,189 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 7)
186	17	Pet households vs pet-friendly sites – Honolulu (wildfire)	Wildfire	3,413 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 7)
187	17	Pet households vs pet-friendly sites – Honolulu (coastal flooding)	Coastal flooding	3,057 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 7)
188	17	Pet households vs pet-friendly sites – Honolulu (tsunami)	Tsunami	1,044 households with a dog or a cat in shelter (usable sites that took pets in 2026 (count): 6)
189	19	Childcare centres in the hurricane zone – Honolulu	Hurricane	263 childcare centres in the zone (childcare centres in the county: 263)
190	19	Childcare centres in the wildfire zone – Honolulu	Wildfire	91 childcare centres in the zone (childcare centres in the county: 263)
191	19	Childcare centres in the tsunami zone – Honolulu	Tsunami	29 childcare centres in the zone (childcare centres in the county: 263)

Rank	Test	Scenario	Hazard	Result
192	19	Childcare centres in the coastal flooding zone – Honolulu	Coastal flooding	19 childcare centres in the zone (childcare centres in the county: 263)
193	19	Childcare centres in the inland flooding zone – Honolulu	Inland flooding	19 childcare centres in the zone (childcare centres in the county: 263)
194	22	Care facilities in the tsunami zone – Honolulu	Tsunami	5 care facilities in the zone (facilities over 15 minutes from a usable site, or none on the island: 2)
195	22	Care facilities in the coastal flooding zone – Honolulu	Coastal flooding	1 care facilities in the zone (facilities over 15 minutes from a usable site, or none on the island: 1)
196	22	Care facilities in the inland flooding zone – Honolulu	Inland flooding	1 care facilities in the zone (facilities over 15 minutes from a usable site, or none on the island: 1)

Questions for counties

These need county knowledge. The plan has no data for them, so it shows no numbers.

1. **Shelter buildings in an earthquake or a strong hurricane:** Which shelter buildings have been assessed for earthquake and hurricane-wind resilience, and which need a check before they are used after a strong event?
2. **Resupply for Molokaʻi and Lānaʻi:** How would water, food, cots, fuel and medical supplies reach Molokaʻi and Lānaʻi if the harbours or airports close for several days, and who decides the order of shipments?
3. **Cell network outages:** Which areas lose mobile phone service first in a long power outage or after wind damage, and how will warnings and shelter information reach them (radio, sirens, door-to-door)?
4. **Schools used as shelters:** When a school is used as a shelter, how long can it stay open before classes must restart, and where do residents go next?
5. **Partner capacity:** Could each county collect partner capacity (cots, meals, staff, vehicles, interpreters) with a county-run form such as Survey123, with no personal data, so the plan can show pledged capacity instead of 'confirm with partner'?